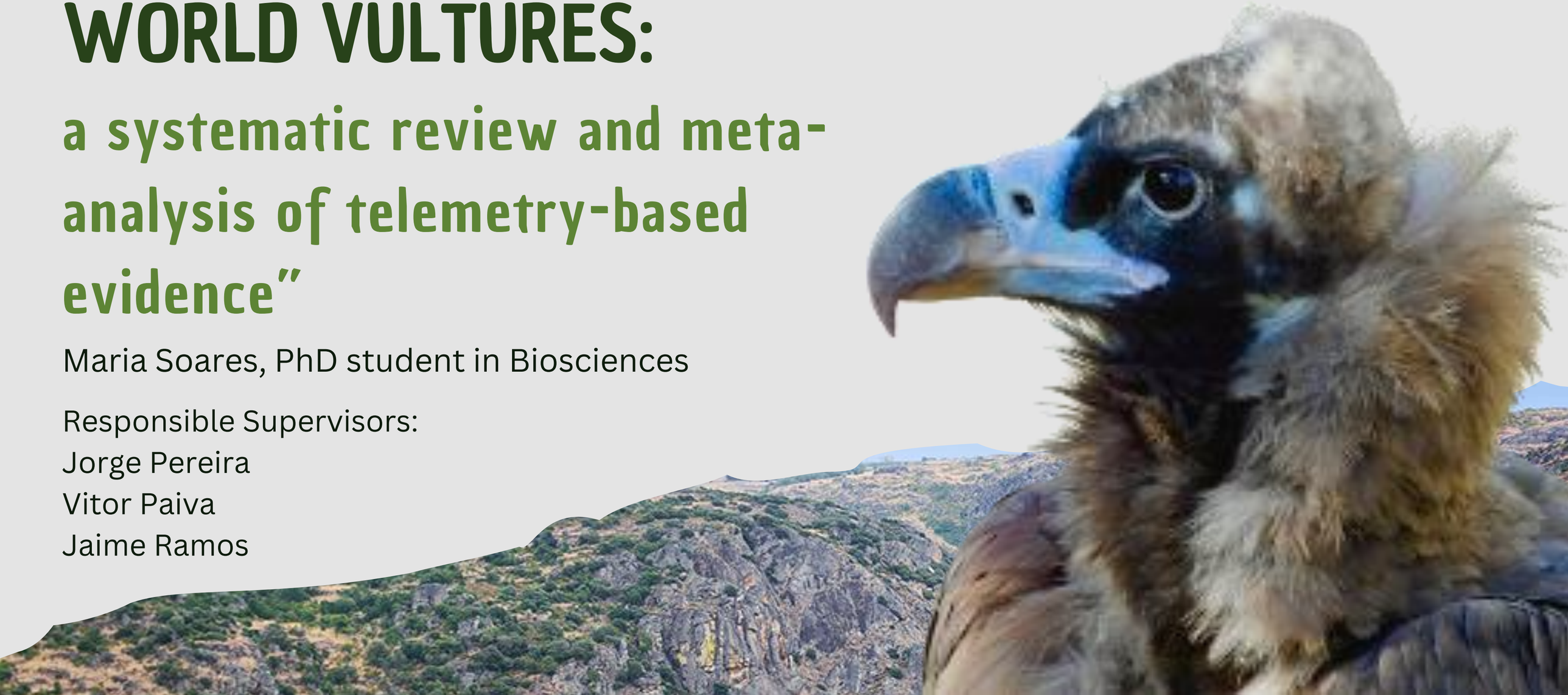
“MOVEMENT AND SPACE USE IN OLD WORLD VULTURES:

a systematic review and meta-analysis of telemetry-based evidence”

Maria Soares, PhD student in Biosciences

Responsible Supervisors:
Jorge Pereira
Vitor Paiva
Jaime Ramos



VULTURES ECOLOGICAL ROLE AND CONSERVATION STATUS

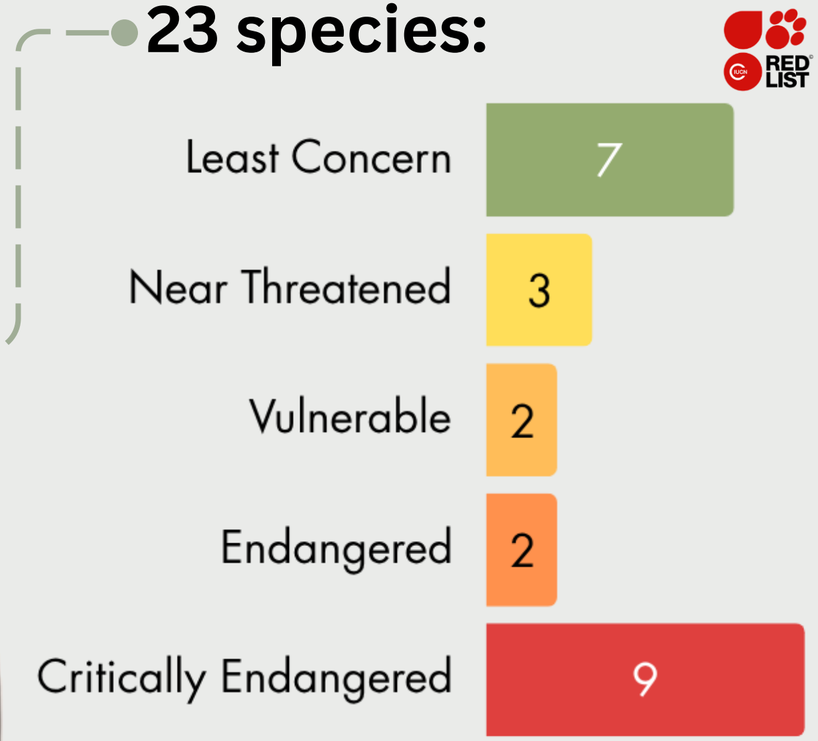
Obligate Scavengers

- Ecologic
- Sanitary
- Economic
- Cultural (...)



© Koshy Koshy

23 species:



Buechley & Şekercioğlu, 2016; Carucci et al., 2022; IUCN, 2024

ENHANCE VULTURES' CONSERVATION EFFORTS

Problem

- Poisoning
- Direct persecution
- Reduced food availability
- Habitat destruction and human disturbance
- Collision/eletrocution with energy infrastructures



© The Urban Birder

Possible solution

Multi-species Action Plan to Conserve African-Eurasian Vulture:

- Objective 11: To support vulture conservation through cross-cutting actions that contribute to addressing knowledge gaps

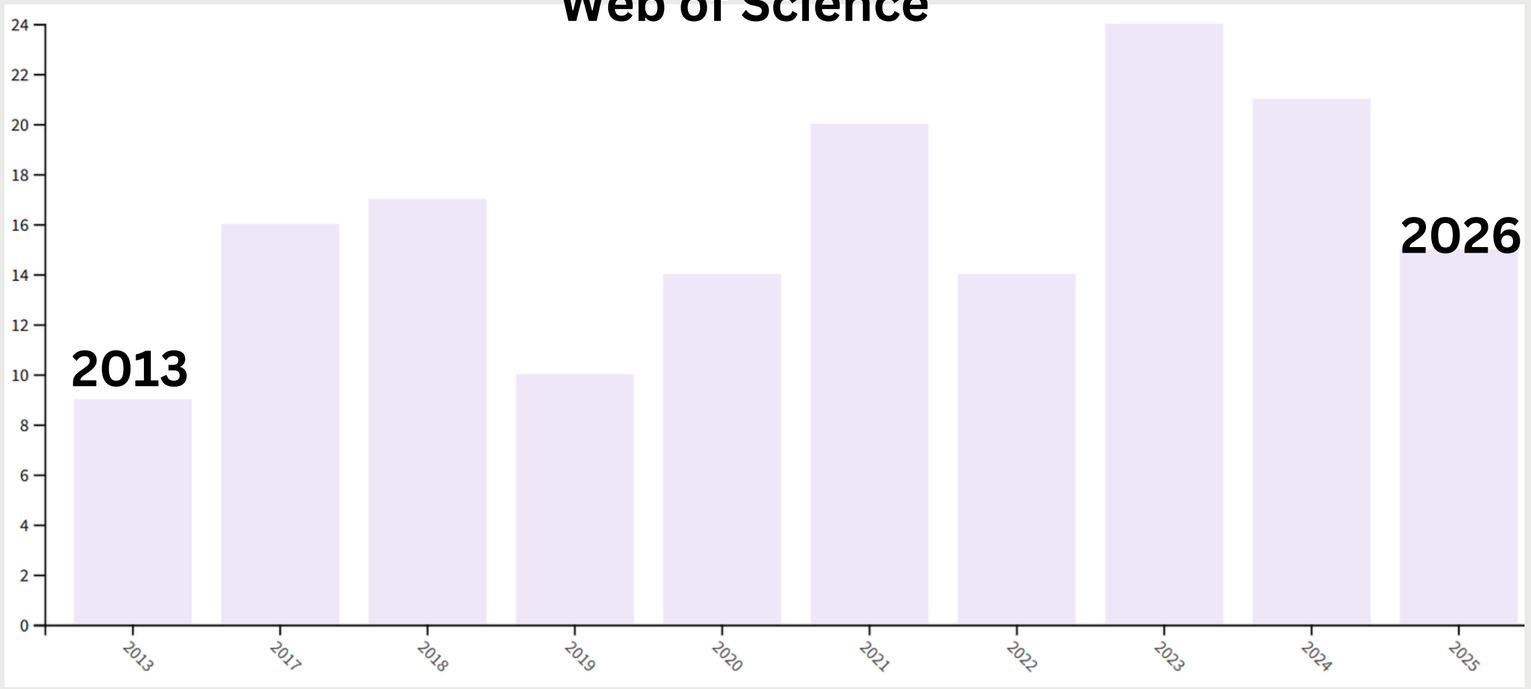
Spatial Ecology

Buechley & Şekercioğlu, 2016; IUCN, 2024; Allen & Singh, 2016; Botha et al. 2017

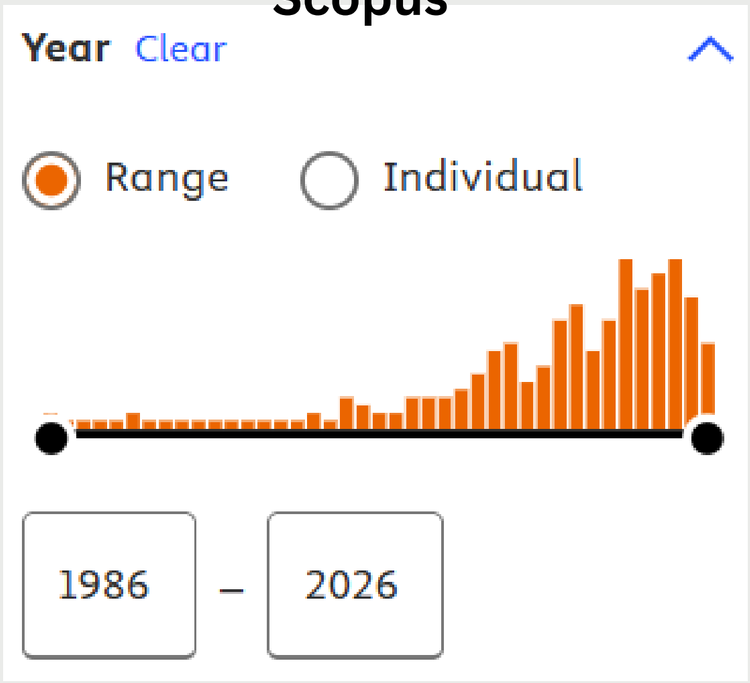
Research on Vultures' Spatial Ecology

(Vulture*) AND (Movement* OR "Movement ecology" OR "Movement patterns" OR "movement behavio*" OR "Spatial ecology" OR "Spatial temporal movement*" OR "Spatial and temporal movement" OR "Space use" OR Rang*) AND ("GPS-tracking" OR GPS* OR "Satellite tracking" OR track* OR "Radio-track*" OR VHF OR logger* OR biologger* OR telemetry)

Web of Science



Scopus



1986 2026

Alarcón & Lambertucci, 2018

Alarcón and Lambertucci *Movement Ecology* (2018) 6:13
https://doi.org/10.1186/s40462-018-0133-5

Movement Ecology

REVIEW Open Access

A three-decade review of telemetry studies on vultures and condors

Pablo A. E. Alarcón[1,2*] and Sergio A. Lambertucci[1]

Summarized telemetry studies on vultures and condors published up to 2017 (compillation of features of existing studies and discussion of the findings under an Movement Ecology Paradigm conceptual framework).

Merely descriptive and narrative

Update needed from 2018 onwards

What should be compilled?

Reported values describing movement patterns

Drivers of variation?

Demographic drivers (age, sex, breeding status)

Seasonal drivers (climatic seasons, breeding season)

- inconsistencies in reported findings
- how movement varies across species, demographic groups and geographical regions
- gaps in the available evidence.

Systematic review and meta-analysis

Main goal (until now): Understand what are the movement and space use patterns, drivers/sources of variations/potential moderators, biases/evidence limitations and conservation applications of telemetry-based spatial ecology research on Old World vultures

“Movement and space use in Old World vultures: a systematic review and meta-analysis of telemetry-based evidence”

Systematic review and meta-analysis

Layer	Main Goal
1: Systematic review (evidence mapping)	Map the geographic, taxonomic and methodological distribution of the quantitative evidence on movement patterns.
2: Meta-analysis	Quantitatively synthesize comparable movement metrics and investigate predefined drivers/sources of variation where sufficient comparable evidence exists.
3: Publication Bias	Asses if the methodological biases or uncertainties omission affect the data viability.
4: Gap Analysis	Identify sampling biases (species, regions, age classes, etc.) in relation to the IUCN status and MsAP.
5: Conservation Application (Stand-by)	Understand whether knowledge has been translated into conservation/what and how can the evaluated and reported values inform conservation actions

Systematic review and meta-analysis

Stage: Between research question formulation, possibly pre-registration and initial search scoping

Possible Framework:

- **P:** Old World Vultures
- **E:** Demographic (age, sex, breeding status) and seasonal (climatic seasons, breeding vs non breeding season)
- **C:** Alternative demographic/seasonal conditions (Comparisons within sexes, comparisons within age classes, within seasons, etc.)
- **O:** Spatial movement metrics (comparable enough within groups/studies)

Thank You!

Aknowledgments:

VCF VULTURE CONSERVATION FOUNDATION

LIFE AEGYPIUS RETURN

